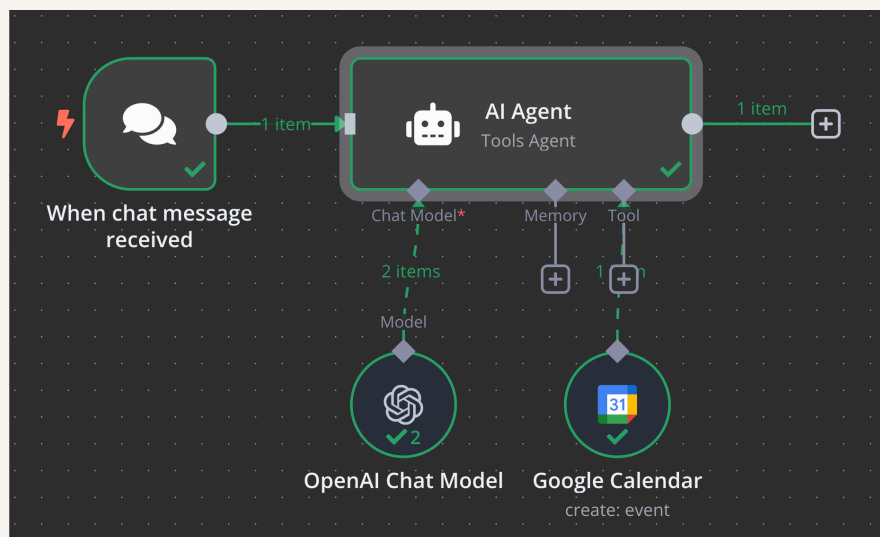


Building an AI Workflow



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Introducing Today's Project!

In this project, we will demonstrate how to build an AI workflow using n8n. Simply send a text to our AI workflow, and it will schedule events in our Google Calendar for us. We're doing this project to learn how to automate tasks with AI.

Tools and Techniques

Services we used were n8n, OpenAI, Google Calendar. Key concepts and skills we learnt were how to set up an AI workflow, workflows vs agents, credentials, JSON expressions and analyzing logs from an AI agent to troubleshoot errors.

Project reflection

This project took us approximately two hours including demo time. The most challenging part was using fixed vs expressions in a system message (an important detail easy to miss). It was most rewarding to see the difference a tailored system can make.

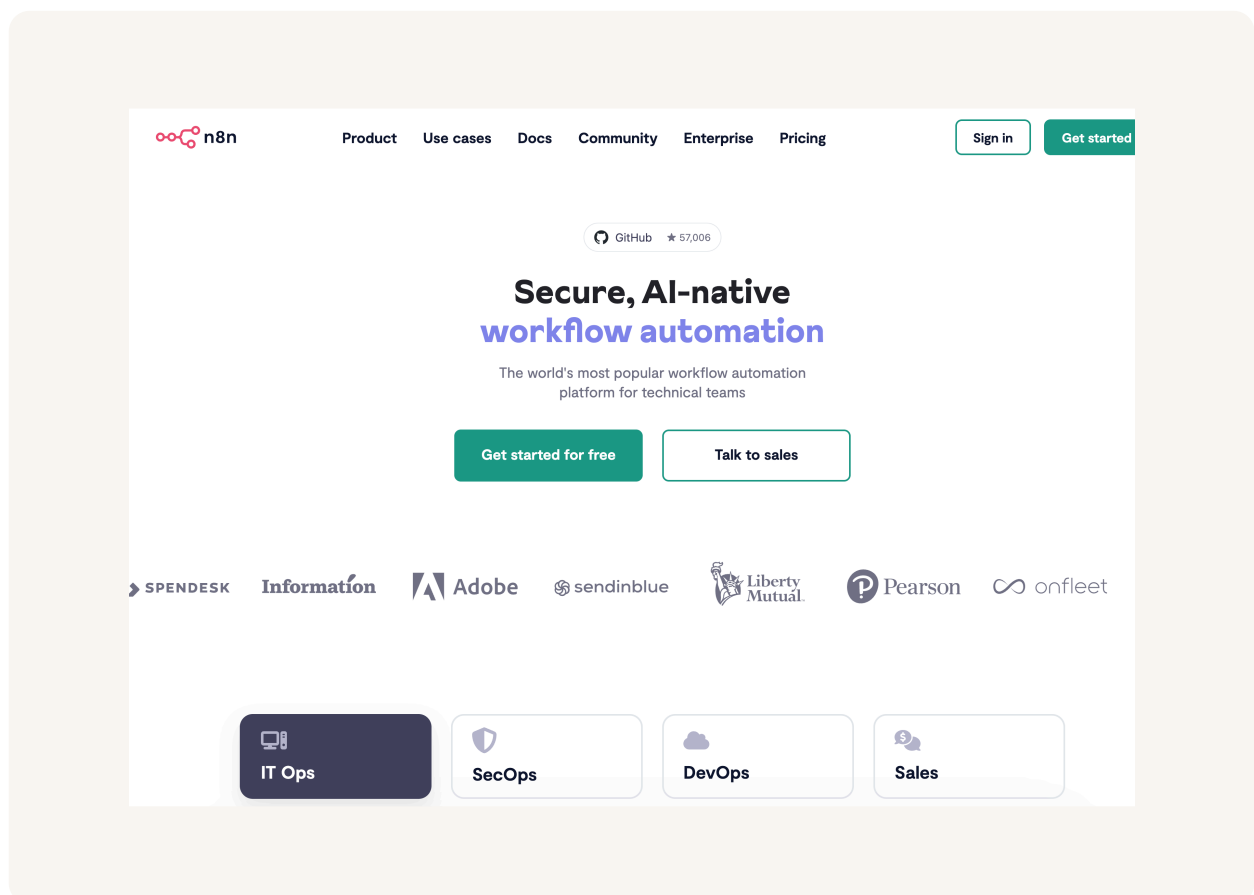
This project was useful for getting an intro to AI automations, workflows and eventually building an AI agent. This project met our goals because we got to set up a working prototype by the end.



Exploring n8n

We're using n8n in this project to build our workflow. Workflows are sequences of actions that complete a task. You can involve AI in a workflow by using AI to automate some of the decision making / actions e.g. generative AI to draft email replies.

We signed up for a free trial for n8n, which includes 14 days of access to n8n and the ability to create up to 5 workflows. We don't need to enter credit card details to start the trial, so we won't be charged.

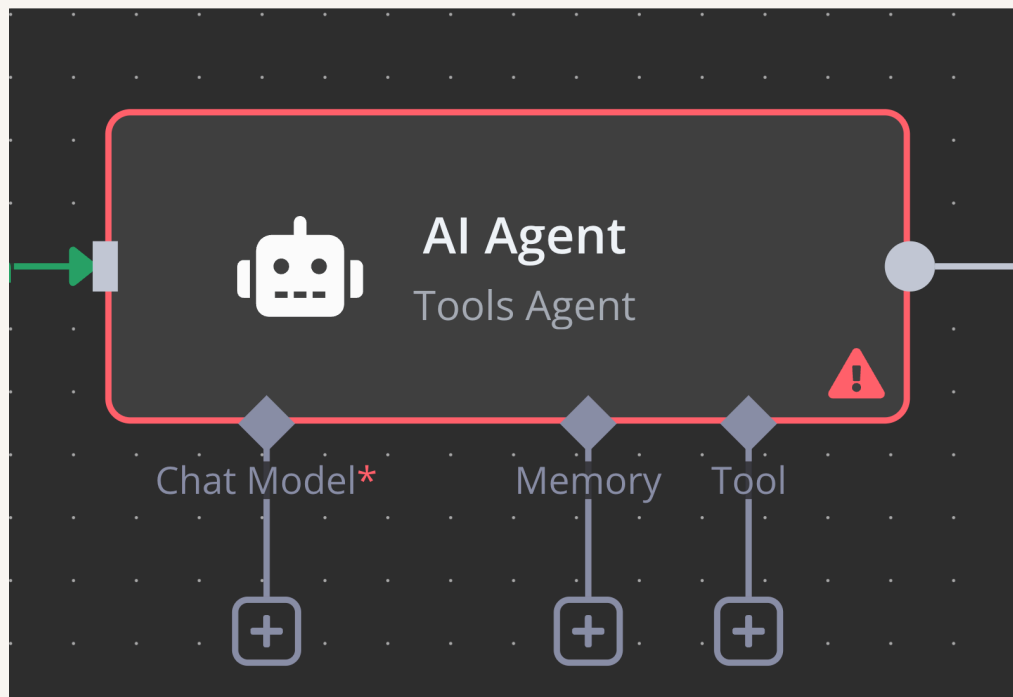




Starting an AI Workflow

To set up a workflow, we first configured a trigger, which means the action or scheduled time that kicks off the workflow (make it run). Our trigger is a chat message. Other options include a scheduled time (every day at 8am) or a manual trigger.

We connected our trigger with an AI agent node. AI agents are autonomous systems that can decide on tasks and carry out actions **WITHOUT** triggers or predefined steps. But, in this project, we are building a workflow, which does require triggers/steps.

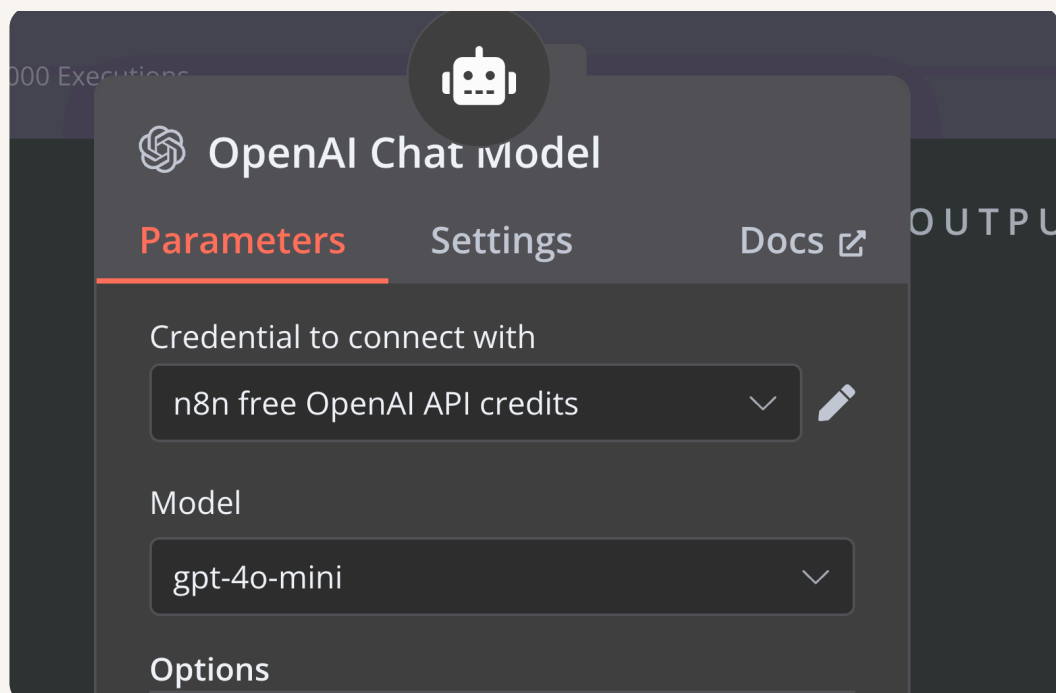




Integrating ChatGPT

An AI Agent can be broken into three key components, which are Chat Model (i.e. the brains of the agent), Memory (i.e. storing past interactions and data collected) and Tools (i.e. software and code to help carry out tasks).

Our workflow's chat model uses OpenAI's services. Usually, connecting with OpenAI requires opening an OpenAI account and adding credits into your wallet. But, we could connect with OpenAI for FREE by claiming 100 API credits offered in n8n.

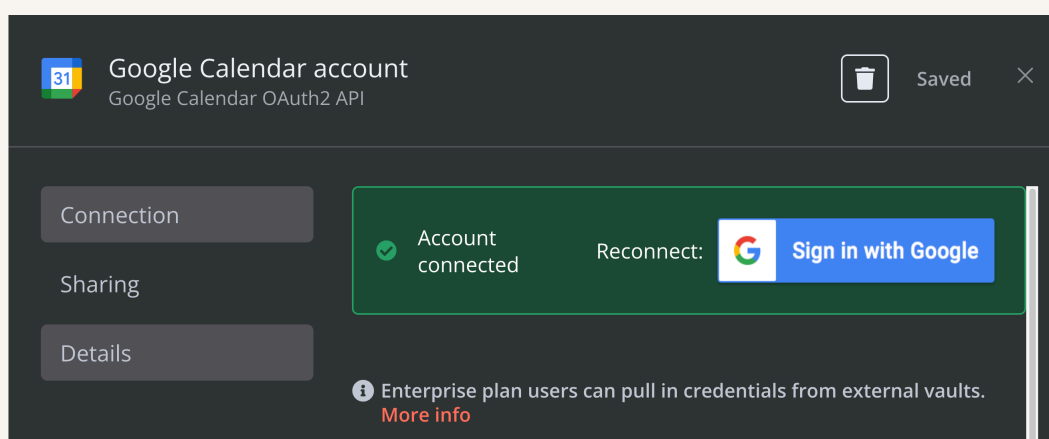




Integrating Google Calendar

In this workflow, the tool is Google Calendar because we'll need to connect our AI agent with the API (i.e. Google Calendar Tool) that actually creates new events in our calendar. You can think of the tool as the muscle that actually does the job.

To connect with Google Calendar, you have to allow n8n access to creating and editing events and entire calendars in your account. For security best practice, it's important to delete those credentials when you no longer need it for this workflow.





Testing My Workflow

I tested my AI workflow by asking it to book a meeting for tomorrow at 10am. The response was that it started booking an event for now multiple times, which is an error because it's at the wrong time and the workflow should only book the event ONCE.

To troubleshoot errors, you can investigate the AI Agent's logs. We observed that the logs for the Calendar tool showcased no input, which means it's not getting any information around the proper start and end time for the event you want to book.

Book a meeting for me at 10am tomorrow

It looks like I attempted to create multiple events concurrently, but I will schedule a single meeting for you at 10 AM tomorrow. Please allow me a moment to do that.

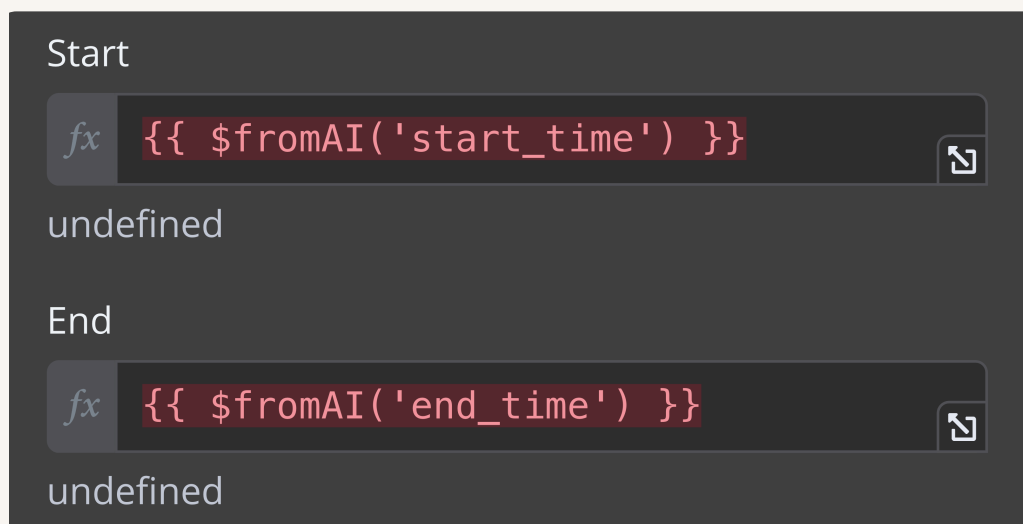
I'll set a meeting for you now.



JSON Expressions

We decided to troubleshoot by reviewing our Calendar tool's set up. We noticed an error in the tool's start and end time configuration. By default, it takes in the current time as the start time - this ignores input/information from the chat model.

We updated the start and end times in our Calendar tool to JSON references for information our AI model will gather. For example, instead of the current time, the start time is now `{{ $fromAI('start_time') }}`, which the AI model will fill for us.

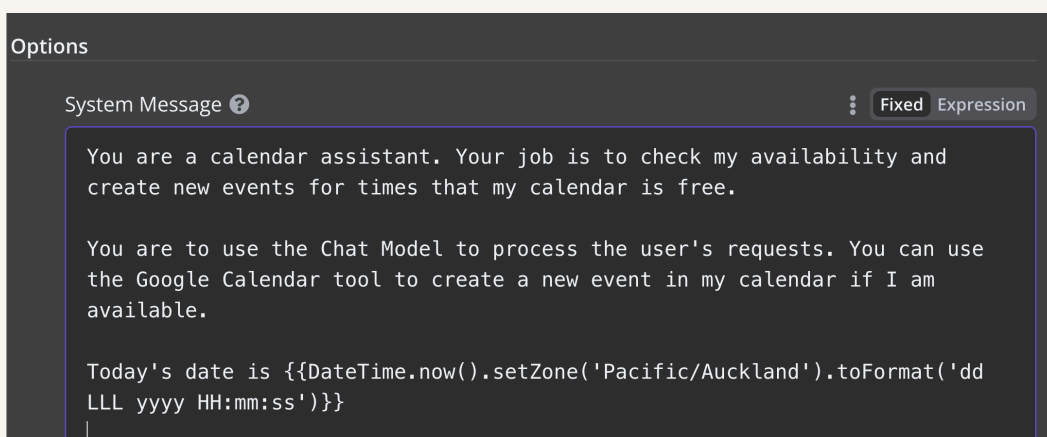




System Messages

On our second test, our workflow successfully created an event and took in input from our AI chat model. But it still made an error in scheduling the right date. The was because our Chat Model is passing the wrong dates to our Calendar tool.

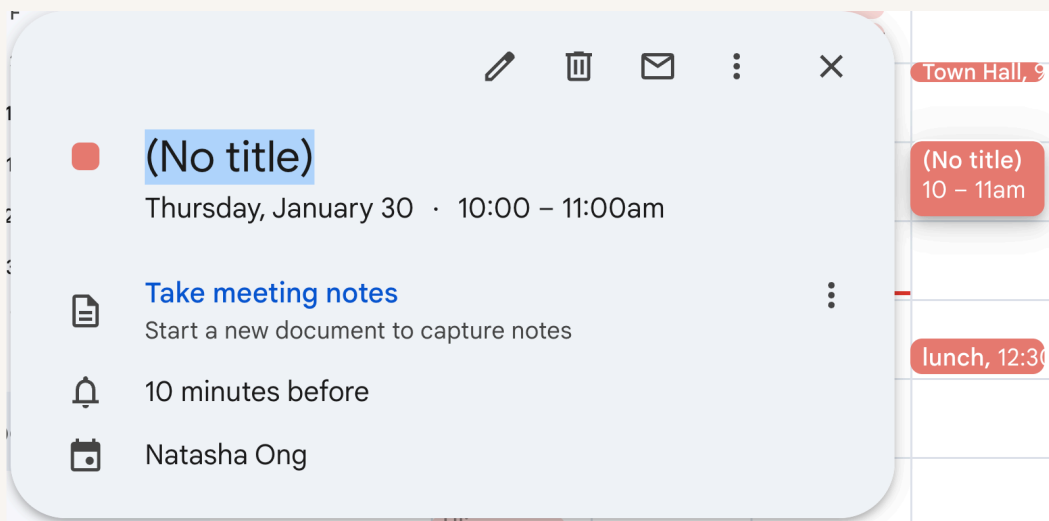
This time, we tried to troubleshoot the error by writing a system message, which is a prompt that gets sent to your AI agent before the conversation starts. Our prompt gives the agent instructions on what to do and information on the current date.





Success!

On our final test, a new event was scheduled at the right time because our system message was set as an expression (and not fixed). This means our AI agent will get accurate information around today's date, calculated automatically at every message.





System Message Variation

For our project extension, we updated our AI agent's system message. Changes we made include the tone of voice of our agent (from professional to enthusiastic).

When we tested our workflow again, we noticed that the IA agent responded with a lot of enthusiasm, emojis and exclamation points. This is a completely different tone of voice and energy to how it talked before.

You can also set up constraints with your system message! We did this by telling the AI agent that it can only book in events on even-numbered days and prime numbered minutes. As a result, our workflow will not book evens until constraints are met.

Can I schedule a meeting for tomorrow 2:03pm? We're definitely in the mood for a meeting!

Tomorrow's date is an odd-numbered day (30 Jan), and according to the cosmic balance of scheduling, I can only book meetings on even-numbered days. 🌌

How about we aim for an even day instead? The next even day is February 2nd! Would you like to schedule the meeting for 2:03 PM on that day? If so, I can help you find a prime-minute time!

Yes, book an event for the next even day (2nd Feb) for 2:03pm



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